Package ‘DATRAS’

June 2, 2026

Type Package

Title Read and convert raw data obtained from
http://datras.ices.dk/Data_products/Download/Download_Data_public.aspx

Version 1.01.1

Date 2026-06-02

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Description Read datras hydrographical data, length data
and age data into single data structure. Convert length
distributions by haul to age distrubutions by haul
using continuation ratio logits.

Suggests mapdata,
maps,
mgcv,
parallel

Imports icesDatras

License GPL

LazyLoad yes

Collate 'ALKfuns.R'
'read_datras.R'
'ices_stuff.R'
'DATRAS.R'
'plotVBG.R'
'downloadExchange.R'
'AgeFunctions.R'

Encoding UTF-8

Config/roxygen2/version 8.0.0

Contents

`DATRAS-package` *Read and convert raw data obtained from ICES*

Description

Read datras hydrographical data, length data and age data into single data structure. Convert length distributions by haul to age distrubutions by haul using continuation ratio logits.

Details

Package:	DATRAS
Type:	Package
Version:	1.0
Date:	2010-06-03
License:	GPL
LazyLoad:	yes

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References

http://datras.ices.dk/Data_products/Download/Download_Data_public.aspx

Examples

```
require(DATRAS)

## Example file
zipfile <- system.file("exchange","Exchange1.zip",package="DATRAS")

## Step 1. Read exchange data into R
d <- readExchange(zipfile)

## Step 2. Preprocess the data
## -Take subset
## -Add size spectrum
d <- subset(d,lon>10,Species=="Gadus morhua")
d <- addSpectrum(d)
d <- addNage(d,2:4)

## Step 3. Convert to data.frame with one line for each response.
df <- as.data.frame(d,response="Nage")
```

---

addNage *Add numbers-at-age to a DATRASraw object.*

Description

Add numbers-at-age to a DATRASraw object. Numbers are estimated using a continuation-ratio logit model. This is just a short-cut for calling 'fitALK' to fit the model followed by 'predict.ALKmodel' and adding numbers-at-age per haul to the DATRASraw object in a variable called 'Nage'.

Usage

```
addNage(
  x,
  ages,
  mc.cores = 1,
  model = c(paste("cra~LngtCm"), paste("cra~poly(LngtCm,2)"),
    paste("cra~LngtCm+s(lon,lat,bs='ts')"))[method],
  method = 1,
  autoChooseK = FALSE,
  useBIC = FALSE,
  varCof = FALSE,
  maxK = 100,
  gamma = 1.4,
  verbose = FALSE,
  ...
)
```

Arguments

`x`	a DATRASraw object.
`ages`	Vector with consecutive ages to consider, last age is plus group.
`mc.cores`	The number of cores to use, i.e. how many processes will be spawned (at most)
`model`	Model formula(string) for ALK, or a vector of strings specifying the formula for each logit (i.e. number of age groups minus one).
`method`	Use default formula: 1=Normal distributed age-at-length (1st order), 2=Second order polynomial, 3=Spatial ALK, first order length-effect.
`autoChooseK`	Automatic choice of the max. dimension for the basis used to represent the smooth term for spatial ALK. See ?s in the mgcv-package.
`useBIC`	Use Bayesian Information Criterion for smoothness selection instead of AIC.
`varCof`	Use varying coefficients model for spatial effect.
`maxK`	Maximum k to use. Only applies if autoChooseK is TRUE.
`gamma`	Multiplier for AIC score (see ?gam)
`verbose`	Print details about the fitting process.
`...`	Optional extra arguments to gam()

Value

A DATRASraw object.

addSpatialData — *Add spatial data from shapefile.*

Description

Add spatial data from a shapefile to a DATRASraw object. Longitude-latitude coordinates from DATRASraw are used to lookup the information in the shapefile. Requires the packages `sf` and `sp`.

Usage

```
addSpatialData(d, shape, select = NULL, ...)
```

Arguments

`d`	DATRASraw object
`shape`	Shapefile or valid filename of existing shapefile.
`select`	Vector of variable names to select from shapefile (default is to select all names)
`...`	Passed to `st_read`

Value

Modified DATRASraw object

addWeightByHaul — *Calculate total biomass by haul and add to HH-records*

Description

Calculate total biomass by haul and add to HH-records

Usage

```
addWeightByHaul(d, to1min = TRUE)
```

Arguments

`d`	DATRASraw object
`to1min`	divide by haul duration in minutes? (defaults to TRUE)

Details

Fits W = a*L^b and applies to all length data (standardized to one minute). Thus, this only makes sense for one species.

Value

DATRASraw object

`AIC.ALKmodel` *Compute Akaike Information Criterion for an age-length key model.*

Description

Compute Akaike Information Criterion for an age-length key model.

Usage

```
## S3 method for class 'ALKmodel'
AIC(object, ..., k = 2)
```

Arguments

`object`	an ALKmodel object.
`...`	unused
`k`	Penalty for number of parameters

Value

AIC value.

`anova.ALKmodel` *Compute approximate likelihood ratio test for the equality of two age-length keys.*

Description

Compute approximate likelihood ratio test for the equality of two age-length keys.

Usage

```
## S3 method for class 'ALKmodel'
anova(object, object2, ...)
```

Arguments

`object`	An object of class 'ALKmodel'
`object2`	An object of class 'ALKmodel'
`...`	unused

Value

a p-value.

`as.data.frame.DATRASraw` *Convert DATRASraw to data.frame suitable for statistical analysis.*

Description

Method to convert DATRASraw to data.frame with one line for each response.

Usage

```
## S3 method for class 'DATRASraw'
as.data.frame(x, ..., format = c("long", "wide"), response, cleanup = TRUE)
```

Arguments

`x`	DATRASraw object to be converted.
`...`	Not used.
`format`	Long or wide format?
`response`	Name of response variable (only needed if data has multiple response variables).
`cleanup`	Remove useless information?

Details

This function should be called as the final step after preprocessing (subset(), addSpectrum(), addNage(), etc). The output format can be controlled by the argument `format`. If format="long" the data.frame has one line for each response which is the standard for most statistical procedures. In the less usual case format="wide" the data.frame has one line for each haul (sample id) with multiple responses across the columns of the response matrix. The wide format thus takes up less memory. If multiple response names are available in `x` (e.g. both "N" and "Nage") then the user must select one through the argument `response`.

Value

data.frame with one line for each response along with associated covariates.

`bayes` *Calculate conditional expectation of some function of length given age.*

Description

Calculate conditional expectation of some function of length given age.

Usage

```
bayes(pal, pl, fl)
```

Arguments

pal	matrix with age-length key, i.e. p(A|L). Columns correspond to age groups.
pl	vector with length distribution, i.e. p(L).
fl	vector of function values at length, i.e. f(L).

Details

Find E(f(L)|A) in terms of p(L), p(A|L) and f(L), using Bayes formula.

Value

vector of expectations

---

bubblePlot — *Plot bubbles on a map with area proportional to "response".*

Description

Plot bubbles on a map with area proportional to "response".

Usage

```
bubblePlot(
  d,
  response = "HaulWgt",
  scale = NULL,
  col.zero = "red",
  pch.zero = "+",
  rim = FALSE,
  ...
)
```

Arguments

d	DATRASraw object
response	name of variable to plot
scale	scale size of bubbles
col.zero	color for zero hauls
pch.zero	pch for zero hauls
rim	Add blue rim to bubbles? (defaults to FALSE)
...	extra arguments to plot

Details

See also addWeightByHaul()

Value

nothing

`c.DATRASraw` *Combine multiple DATRASraw objects*

Description

Method to combine many datasets

Usage

```
## S3 method for class 'DATRASraw'
c(...)
```

Arguments

`...`	DATRASraw objects to put together.

Details

This function combines DATRASraw objects by `rbinding` each of the three components of the objects. The method is useful when downloading large amounts of data in small chunks. If the same haul.id is present within different datasets the method will stop. If a variable name is only present in some of the datasets, the variable will be added to the remaining datasets (filled with NA) and a warning will be triggered.

Value

Merged dataset.

Examples

```
x <- readExchange(file1)
y <- readExchange(file2)
z <- c(x,y)
```

`DATRAS-internal` *Internal DATRAS Functions*

Description

Internal DATRAS functions

Details

These are not to be called by the user (or in some cases are just waiting for proper documentation to be written :).

`downloadExchange` *Interface to DATRAS php download script*

Description

Interface to DATRAS php download script

Usage

```
downloadExchange(survey, years = NULL)
```

Arguments

`survey`	Name of survey to download (For a list of available names run `downloadExchange()`)
`years`	Vector of years to download. If unspecified, all years are downloaded one by one.

Details

Note: Command line php must be installed.

fitALK — *Fit a continuation-ratio logit model for age given length and possibly other covariates.*

Description

Fit a continuation-ratio logit model for age given length and possibly other covariates.

Usage

```
fitALK(
  x,
  minAge,
  maxAge,
  mc.cores = 1,
 model = c("cra~LngtCm", "cra~poly(LngtCm,2)", "cra~LngtCm+s(lon,lat,bs='ts')")[method],
  method = 1,
  autoChooseK = FALSE,
  useBIC = FALSE,
  varCof = FALSE,
  maxK = 100,
  gamma = 1.4,
  verbose = FALSE,
  ...
)
```

Arguments

`x`	a DATRASraw object.
`minAge`	minimum age group to consider
`maxAge`	maximum age group to consider
`mc.cores`	The number of cores to use, i.e. how many processes will be spawned (at most)
`model`	Model formula(string) for ALK, or a vector of strings specifying the formula for each logit (i.e. number of age groups minus one).
`method`	Use default formula: 1=Normal distributed age-at-length (1st order), 2=Second order polynomial, 3=Spatial ALK, first order length-effect.
`autoChooseK`	Automatic choice of the max. dimension for the basis used to represent the smooth term for spatial ALK. See ?s in the mgcv-package.
`useBIC`	Use Bayesian Information Criterion for smoothness selection instead of AIC.
`varCof`	Use varying coefficients model for spatial effect.
`maxK`	Maximum k to use. Only applies if autoChooseK is TRUE.
`gamma`	Multiplier for AIC score (see ?gam)
`verbose`	Print details about the fitting process.
`...`	Optional extra arguments to gam()

Value

An object of class 'ALKmodel'

`fitALKone` *Fit part of the continuation ratio model using GAMs (helper function)*

Description

Fit part of the continuation ratio model using GAMs (helper function)

Usage

```
fitALKone(
  a,
  ages,
  AL,
  model,
  gamma,
  autoChooseK = FALSE,
  useBIC = FALSE,
  varCof = FALSE,
  maxK = 100,
  verbose = FALSE,
  ...
)
```

Arguments

`a`	Condition on being at least this age.
`ages`	Vector with consecutive ages to consider, last age is plus group.
`AL`	Age-length data from a DATRASraw object
`model`	Model formula (string or formula), or a vector of strings specifying the formula for each age group.
`gamma`	Multiplier for AIC score (see ?gam)
`autoChooseK`	Automatic choice of the max. dimension for the basis used to represent the smooth term for spatial ALK. See ?s in the mgcv-package.
`useBIC`	Use Bayesian Information Criterion for smoothness selection instead of AIC.
`varCof`	Use varying coefficients model for spatial effect.
`maxK`	Maximum k to use. Only applies if autoChooseK is TRUE.
`verbose`	Print model summary?
`...`	Extra parameters to gam()

Value

Object of class '"gam"'

`fitvbg` *Fit von Bertalanffy growth model to age-length data.*

Description

Fit von Bertalanffy growth model to age-length data.

Usage

```
fitvbg(d, L0 = 0, plot = FALSE)
```

Arguments

`d`	DATRASraw object
`L0`	Length at time t0
`plot`	Plot the fitted vbg curve?

Details

Note: Fitting is done using non linear least squares which assumes constant variance of size samples given age.

Value

list with parameters

`getDatrasExchange` *Read exchange data into R.*

Description

Download raw exchange data from DATRAS web services.

Usage

```
getDatrasExchange(
  survey,
  years,
  quarters,
  strict = TRUE,
  download.hl = TRUE,
  download.ca = TRUE,
  verbose = TRUE
)
```

Arguments

`survey`	Name of survey to download (For a list of available names run `icesDatras::getSurveyList()`)
`years`	Vector of years to download.
`quarters`	Vector of quarters to download.
`strict`	if TRUE, missing haul ids in age data should be unqiuely matched when filled in, if FALSE a random match will be assigned.
`download.hl`	if FALSE, length frequency data in DATRAS ("HL") are not downloaded. Default: TRUE
`download.ca`	if FALSE, biological samples in DATRAS ("CA") are not downloaded. Default: TRUE
`verbose`	Print information? Default: TRUE

Details

This function downloads raw exchange data and converts it to a DATRASraw object. If multiple surveys are supplied, all files are read and combined to a single object.

Value

DATRASraw object.

`icesSquare` *Longitude/Latitude to ICES square conversion.*

Description

Calculate ICES square from Longitude/Latitude pairs.

Usage

```
icesSquare(data, lon = data$lon, lat = data$lat)
```

Arguments

`data`	A DATRASraw object or a data.frame with Longitude/Latitude information.
`lon`	Vector of longitudes.
`lat`	Vector of latitudes.

Value

Vector of ICES square names.

icesSquare2coord — *ICES square to Longitude/Latitude conversion.*

Description

Calculate Longitude/Latitude pairs from ICES square.

Usage

```
icesSquare2coord(x, format = c("corner", "midpoint", "polygons"))
```

Arguments

x	Vector of ICES square names.
format	Return points or polygons?

Details

Output from this function depends on argument `format`:

1. format="corner": Return south-west corner of square.
2. format="midpoint": Return midpoint of square.
3. format="polygons": Return list of square polygons (suitable for `polygon` plot function).

Value

data.frame with coordinates or list of polygons.

Examples

```
icesSquare2coord(c("37G1","37G2"))
icesSquare2coord(c("37G1","37G2"),"polygons")
```

minus9toNA — *Replace -9 with NA in numeric columns*

Description

Replace -9 with NA in numeric columns

Usage

```
minus9toNA(x)
```

Arguments

`x`	DATRASraw object

Value

DATRASraw object

`NageByHaul` *Calculate numbers-at-age for a particular haul.*

Description

Calculate numbers-at-age for a particular haul.

Usage

```
NageByHaul(row, x, returnALK = FALSE)
```

Arguments

`row`	Row number (haul) to be predicted.
`x`	An object of class "ALKmodel"
`returnALK`	Return ALK instead of numbers-at-age

Value

Vector with numbers-at-age (or ALK)

`plot.DATRASraw` *Plot map with haul positions.*

Description

Plot map with haul positions and ICES squares. If a response variable is present also plot frequencies within each square.

Usage

```
## S3 method for class 'DATRASraw'
plot(
  x,
  add = FALSE,
  pch = 16,
  plot.squares = !add,
  plot.map = !add,
  plot.points = TRUE,
  plot.response = TRUE,
  xlim = NULL,
  ylim = NULL,
  ...
)

## S3 method for class 'DATRASraw'
points(x, ...)
```

Arguments

`x`	DATRASraw object.
`add`	Plot on top of existing plot?
`pch`	Point type of positions.
`plot.squares`	Add ICES squares?
`plot.map`	Add a map?
`plot.points`	Plot haul positions?
`plot.response`	Plot frequencies of response if present?
`xlim`	Longitude range of plot.
`ylim`	Latitude range of plot.
`...`	Controlling plot of positions.

Examples

```
x <- readExchange(file1)
y <- subset(x,lon>9,LngtCm<60)
plot(y,col="red")
##' Add response variable
y <- addSpectrum(y)
plot(y,col="red")
```

plotALKfit *plot fitted ALK for a given haul*

Description

plot fitted ALK for a given haul

Usage

```
plotALKfit(x, row, lwd = 2, type = "l", ylab = "Proportion", xlab = "Cm", ...)
```

Arguments

`x`	An object of class 'ALKmodel'
`row`	row number in hydro data (i.e. haul number)
`lwd`	argument to matplot()
`type`	argument to matplot()
`ylab`	argument to matplot()
`xlab`	argument to matplot()
`...`	extra parameters for matplot()

Value

nothing

plotALKraw *Plot a raw ALK using the observed proportions in each age group.*

Description

Plot a raw ALK using the observed proportions in each age group.

Usage

```
plotALKraw(
  x,
  minAge,
  maxAge,
  truncAt = 0,
  type = "l",
  ylab = "Proportion",
  xlab = "Cm",
  ...
)
```

Arguments

`x`	a DATRASraw object with added spectrum.
`minAge`	pool all ages less than or equal to this age.
`maxAge`	pool all ages greater than or equal to this age.
`truncAt`	truncate proportions below this number for nicer plots
`type`	argument to matplot()
`ylab`	argument to matplot()
`xlab`	argument to matplot()
`...`	extra parameters to matplot()

Value

nothing

`plotVBG` *Cohort plot*

Description

Plot size-spectra with von Bertalanffy growth trajectories.

Usage

```
plotVBG(
  d,
  model = fitvbg(d),
  by = paste(Year, Quarter),
  scale = 1,
  xlim = c(min(d$abstime) - 1, max(d$abstime)),
  ylim = c(0, max(d[["HL"]]$LngtCm, na.rm = TRUE)),
  col = grey(0.9),
  yearClasses = (floor(xlim[1]) - 10):ceiling(xlim[2]),
  mar = c(0, 0, 0, 0),
  oma = c(5, 5, 5, 5),
  ylab = "Length (cm)",
  ...
)
```

Arguments

`d`	DATRASraw objects to put together.
`model`	list containing von Bertalanffy growth parameters. By default a simple fit will be performed based on the data.
`by`	Grouping variable to define the individual size-spectra.

scale	Control height of spectra.
xlim	Time range in years.
ylim	Size range in cm.
col	Color of the histograms. Recycled if shorter than number of histograms.
yearClasses	Growth curves are shown for these year classes.
mar	Controls plot margin.
oma	Controls plot outer margin.
ylab	y-axis label
...	passed to par()

Details

This function draws empirical size distributions of a DATRASraw object in order to follow cohorts using the von Bertalanffy growth model.

Examples

```
d <- addSpectrum(d)
plotVBG(d)
```

---

predict.ALKmodel *Predict method for ALKmodel objects*

Description

Predict method for ALKmodel objects

Usage

```
## S3 method for class 'ALKmodel'
predict(object, newdata = NULL, type = "Nage", mc.cores = 1, ...)
```

Arguments

object	An object of class 'ALKmodel'
newdata	optionally, a DATRASraw object to predict
type	the type of prediction required. The default is "Nage", which is numbers-at-age for each haul. The other option is "ALK", which gives a list of age-length keys, one for each haul.
mc.cores	use this number of cores (parallel computation via `parallel` library)
...	unused

Value

A matrix if type equals "Nage", and a list of matrices if type equals "ALK".

rawALK — *Calculate empirical (raw) age length key.*

Description

Calculate empirical (raw) age length key.

Usage

```
rawALK(d, minAge, maxAge)
```

Arguments

`d`	DATRASraw object
`minAge`	Minimum age
`maxAge`	Maximum age (plus group)

Details

Ages < minAge are excluded. Ages > maxAge are pooled into a plus group.

Value

matrix with ALK

readExchange — *Read exchange data into R.*

Description

Unpack and read raw exchange zipfile(s).

Usage

```
readExchange(zipfile, strict = TRUE)
```

Arguments

`zipfile`	File(s) to read.
`strict`	if TRUE, missing haul ids in age data should be unqiuely matched when filled in, if FALSE a random match will be assigned.

Details

The raw exchange files from ICES are zipfiles (without zip extension). This function unzips the file to a temporary directory and then reads and converts the resulting text file. If multiple filenames are supplied, all files are read and combined to a single object.

Value

DATRASraw object.

---

readExchangeDir *Read all exchange files in a folder.*

Description

Read all exchange files in a folder and combine to a single DATRASraw object

Usage

```
readExchangeDir(path = ".", pattern = ".zip", strict = TRUE)
```

Arguments

path	File path.
pattern	Pattern that the exchange files match.
strict	if TRUE, missing haul ids in age data should be unqiuely matched when filled in, if FALSE a random match will be assigned.

Details

Calls `readExchange` on all the exchange files in the folder and combines the results.

Value

DATRASraw object

renameDATRAS — *Rename DATRAS columns*

Description

Rename some columns to the old names

Usage

```
renameDATRAS(x)
```

Arguments

`x`	part of a DATRASraw object (CA, HH or HL)

Value

part of a DATRASraw object (CA, HH or HL) with renamed columns

subset.DATRASraw — *Subsetting a DATRASraw object.*

Description

General method for subsetting a DATRASraw object.

Usage

```
## S3 method for class 'DATRASraw'
subset(x, ..., na.rm = TRUE)
```

Arguments

`x`	Take subset of this dataset.
`...`	One or more subset criteria.
`na.rm`	Discard missing values in subset criterion?

Details

A DATRASraw object contains two kinds of information:

1. Between haul information (contained in component 2)
2. Within haul information (contained in component 1 and 3)

This method can subset over both kinds of information - at once. The functionality is achieved by matching each of the subset critera to the appropriate data component. When a match is found the subset is performed on the component and the subsetting is continued to the next subset criteria. The number of subset critera can be arbitrary and is given through the `...` argument.

Value

The reduced dataset.

Examples

```
x <- readExchange(file1)
y <- subset(x,Species=="Gadus morhua",LngtCm>30,HaulDur>25)
```

`tilePlot` *Add small plot on existing plot.*

Description

Add new plot on existing plot.

Usage

```
tilePlot(expr, xlim, ylim, pol)
```

Arguments

`expr`	Expression of plot commands producing new plot.
`xlim`	New plot x-range.
`ylim`	New plot y-range.
`pol`	Polygon defining the sub-window wrt. large plot coordinate system.

Details

Add new plot on existing plot.

Value

Output of evaluated `expr`

`weightAtAge` *Calculate weight by age group in the survey accounting for length-stratified subsampling of age.*

Description

Calculate weight by age group in the survey accounting for length-stratified subsampling of age.

Usage

```
weightAtAge(d, minAge, maxAge, alk.type = c("raw", "smooth"))
```

Arguments

`d`	DATRASraw object
`minAge`	minimum age group to consider
`maxAge`	maximum age group (plus group)
`alk.type`	method to calculate age-length key

Details

Empirical application of Bayes formula.

Value

A vector with weight-at-age

`[.DATRASraw` *Sample unit subset.*

Description

Sample unit subset for DATRASraw object.

Usage

```
## S3 method for class 'DATRASraw'
x[i]
```

Arguments

`x`	DATRASraw object
`i`	Integer vector

Details

Extract all information related to a given set of hauls. For instance x[1:2] extracts the first two hauls. Duplicated integer values results in automatic renaming of haul ids. This can be useful for sampling with replacement.

Value

DATRASraw object

Examples

```
x[1:2]
x[sample(3,replace=TRUE)]
split(x,x$Country)
```

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